

01 Sets (Set Concepts)

OPENING VISUAL

How many stones are in my hand?

Think alone

Pair talk

Share one idea

Today's mission

Goal: I can distinguish between finite and infinite sets using

1 Learn

Use: Can We Count Everything?, Defining Our Sets

2 Practise

Show: answer with one reason or example.

3 Show

Routine: listen, talk, try, explain.

4 Reflect

Explain in your own words.



Starter: think first

1

THINK

**How many stones are
in my hand?**

30 seconds

2

PAIR

**compare your first
answer with a partner.**

with partner

3

SHARE

**connect your answer
to Can We Count
Everything?.**

whole class

Can We Count Everything?

CONTENT MAP

Concrete exploration of countable objects

Practical
counting of
stones

Attempting to
count grains of
sand

Identifying
which count
can be
finished

Defining Our Sets

MODEL MAP

Worked example

Finite Set: Has a
countable
number of
members

Examples from
daily life

Infinite Set: Has
members that go
on forever

Try it together

PRACTICE FLOW

Answer with evidence

?

Can you tell me exactly
how many grains are in
that pile of sand?

answer + reason

?

Which count was easier
to finish?

answer + reason

?

What is a finite set?

answer + reason

Use evidence from the lesson, not guessing.

Activity challenge

GROUP INFOGRAPHIC

Challenge

In groups of five,
learners count a
pile of 10

Draw two Venn
diagrams. In
the first, they

Write the
definitions of
finite and infinite

Share-out: explain one result, then improve it.

Exit ticket

EXIT TICKET

Answer: Can a very large set still be finite? (Yes, if it has an end).

Use one complete sentence.

Use one key word: Set

Hand in or read your answer aloud.